

Quaia Redshift Dipole and Quadrupole Analysis (0V1)

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1. Executive Summary

This report summarises the current Quaia-based analysis of large-scale anisotropy in the mean quasar redshift field. Using the Gaia–unWISE Quaia catalogue ($G < 20$), we construct HEALPix maps of the mean redshift $\langle z \rangle$, fit a dipole, and then study the residual field and its quadrupole ($\ell = 2$) component. A suite of random mocks based on the Quaia random catalogue and selection function is used to estimate the expected noise level and to assess the significance of the observed signals.

Key points:

- The mean redshift field shows a clear dipole aligned close to the Galactic anti-centre, with fractional amplitude $\Delta z / z \approx 6.0 \times 10^{-3}$, corresponding to $\approx 4.5\sigma$ above selection + shot-noise mocks.
- After subtracting the best-fit dipole, the residual mean-redshift field displays a very strong quadrupole whose symmetry axis is almost aligned with the North Galactic Pole. The quadrupole power at $\ell = 2$ remains roughly two orders of magnitude above the mock expectation even when restricting to $|b| > 30^\circ$ and applying the official selection function.

- The combination of a strong dipole and a strong quadrupole indicates that the distance–redshift relation traced by quasars is not isotropic on the sky, at least under the assumptions of the mock model. Whether this is cosmological (e.g. anisotropic metric / $H(z)$ / variable- c) or due to subtle survey systematics requires further work.
- These findings are qualitatively consistent with earlier P_{Λ}/FR analyses of BAO data, where anisotropic DH/DM ratios (with $DH/DM \approx 0.35$ along a preferred axis) suggested differential radial vs transverse expansion. Quaia provides an independent quasar-based probe of similar large-scale structure in the STF (space-time fabric).

2. Acronyms and Notation

Quaia : Gaia–unWISE quasar catalogue (version 0.1.0, $G < 20.0$ used here).

HEALPix : Hierarchical Equal Area isoLatitude Pixelisation of the sphere.

NSIDE : HEALPix resolution parameter ($n_{\text{pix}} = 12 \times \text{NSIDE}^2$).

ℓ, m : Spherical harmonic multipole indices.

C_ℓ : Angular power spectrum coefficients of a field on the sphere.

D_ℓ : $\ell(\ell+1) C_\ell / (2\pi)$, used for visualising angular power.

$\langle z \rangle$: Mean redshift in a given sky pixel.

Dipole : $\ell = 1$ component; one-lobed pattern (one hemisphere high, one low).

Quadrupole: $\ell = 2$ component; two-lobed / ring vs poles pattern.

(l, b) : Galactic longitude and latitude (degrees).

GC : Galactic Centre ($l = 0^\circ, b = 0^\circ$).

anti-GC : Galactic anti-centre ($l = 180^\circ, b = 0^\circ$).

BAO : Baryon Acoustic Oscillations.

DH : BAO radial distance measure ($\propto 1 / H(z)$).

DM : BAO transverse comoving distance measure.

DH/DM : Ratio of radial to transverse BAO distances.

FR : Full Relativity cosmological framework in PLamb analysis.

STF : Space-Time Fabric (PLamb terminology for large-scale metric behaviour).

3. Datasets Used

3.1 Quiaia quasar catalogue ($G < 20$)

Primary quasar catalogue:

- quiaia_G20.0.fits (main quasar sample with $G < 20.0$)
- random_G20.0_10x.fits (random angular positions at $10\times$ density)
- selection_function_NSIDE64_G20.0.fits (NSIDE = 64 selection function map)

For the present analysis the essential columns are:

- ra : Right Ascension (ICRS, degrees)
- dec : Declination (ICRS, degrees)
- redshift_quaia : photometric redshift estimate (used as z)

3.2 Reduced CSV catalogue

To simplify the analysis pipeline, a reduced CSV was created:

- /Users/boyde/Downloads/quiaia_G20p0_basic.csv

Columns:

- RA : Right Ascension (degrees, ICRS)
- DEC : Declination (degrees, ICRS)
- Z : redshift_quaia

Galactic coordinates were added via `qso_add_galactic_coords.py`, giving:

- `/Users/boyde/Downloads/quaia_G20p0_basic_gal.csv`

Additional columns:

- `L_GAL` : Galactic longitude (degrees)
- `B_GAL` : Galactic latitude (degrees)

3.3 Random catalogue and selection function

The Quaia random catalogue and selection function are used to:

- 1) Define a selection-aware mask ($\text{sel} \geq \text{sel_min_frac} \times \max(\text{sel})$, typically 0.3).
- 2) Build mocks by drawing random positions from `random_G20.0_10x.fits` and assigning redshifts by resampling the real Quaia redshift distribution.

4. Methodology

4.1 HEALPix mean-redshift maps

Using `NSIDE = 64`, each quasar is converted from (RA, DEC) to Galactic (l, b)

and then to a HEALPix pixel p . For each pixel we accumulate:

- N_p : number of quasars in pixel p
- Σz_p : sum of redshifts in pixel p
- $\langle z \rangle_p : \Sigma z_p / N_p$ (if $N_p > 0$; otherwise pixel is set to UNSEEN)

This yields:

- `counts_map` N_p
- `mean_z_map` $\langle z \rangle_p$

4.2 Redshift dipole model

The first-order anisotropy model for the mean- z field is:

$$\langle z(\hat{n}) \rangle \approx z_0 + D \cdot \hat{n}$$

where

z_0 : global mean redshift

D : 3D dipole vector (units of redshift)

$\hat{n}$: unit vector in the direction (l, b)

For pixel p we define

$$\delta z_p = \langle z \rangle_p - z_0 \approx D \cdot \hat{n}_p,$$

and solve for D by weighted least squares, with weights $\propto N_p$.

4.3 Pixel selection

To ensure a robust fit, pixels are required to satisfy:

- $N_p \geq N_{\min}$ (commonly $N_{\min} = 10$)
- selection function $\text{sel}_p \geq \text{sel}_{\min_frac} \times \max(\text{sel})$; typically $\text{sel}_{\min_frac} = 0.3$

This excludes sparsely populated pixels and regions with poor completeness.

4.4 Residual map and quadrupole

Once D and z_0 are determined, we construct a dipole model map:

$$z_{\text{model}}(\hat{n}) = z_0 + D \cdot \hat{n}$$

The residual map is

$$z_{\text{res}}(\hat{n}) = \langle z(\hat{n}) \rangle - z_{\text{model}}(\hat{n})$$

We then compute the angular power spectrum C_ℓ of $z_{\text{res}}(\hat{n})$ using `healpy.anafast`, optionally with a $|b|$ cut (e.g. $|b| \geq 30^\circ$) in addition to the selection mask.

From C_ℓ we define

$$D_\ell = \ell(\ell + 1) C_\ell / (2\pi)$$

and examine particularly $\ell = 2$ (quadrupole). A strong peak at $\ell = 2$ implies a strong quadrupole in the residual mean- z field.

To visualise the quadrupole pattern, we can project the residual field onto spherical harmonics up to $\ell = 2$, set all modes except $\ell = 2$ to zero, and reconstruct the corresponding sky map. This isolates the quadrupole component and its axis.

4.5 Random mocks and significance

Random mocks are generated as follows:

- 1) Draw N_{real} random positions from `random_G20.0_10x.fits` (with replacement if needed).
- 2) Draw N_{real} redshift values by resampling the real Quia redshift array (with replacement).
- 3) Build mean- z HEALPix maps, fit the dipole, and compute the residual map and C_ℓ

exactly as for the real data.

From each mock we record:

- $\text{dipole_frac} = |D| / z_0$
- $D2_residual = D\ell$ at $\ell = 2$ (quadrupole of residual map)

The distribution of these quantities across nmocks provides the expected level of fluctuations from the mask, selection, and shot noise. Comparing the real values to this distribution yields an approximate “ σ -level” significance under this null model.

5. Results

5.1 Global redshift dipole

For Quiaia $G < 20$, $NSIDE = 64$, $\text{sel} \geq 0.3 \times \max(\text{sel})$, and $N_p \geq 10$, the dipole fit yields:

- $N_{\text{objects}} \approx 755\,850$
- $z_{\text{global_mean}} \approx 1.5135$
- $|D| \approx 9.16 \times 10^{-3}$
- $\text{dipole_frac} \approx 6.05 \times 10^{-3}$
- $l_{\text{dip}} \approx 180.5^\circ$ (Galactic longitude)
- $b_{\text{dip}} \approx 13.7^\circ$ (Galactic latitude)

Thus the mean quasar redshift is about 0.6% higher in the direction

$(l, b) \approx (180.5^\circ, 13.7^\circ)$ (near the Galactic anti-centre) than in the opposite direction.

5.2 Dipole significance from mocks

A suite of 50 mocks with the same NSIDE, selection cut, and N_p threshold yields:

- $\langle \text{dipole_frac} \rangle_{\text{mocks}} \approx 2.07 \times 10^{-3}$
- $\sigma(\text{dipole_frac})_{\text{mocks}} \approx 8.89 \times 10^{-4}$

The observed value then corresponds to

$$z_{\text{dip}} \approx (6.05 \times 10^{-3} - 2.07 \times 10^{-3}) / (8.89 \times 10^{-4}) \approx 4.5 \sigma$$

Within the assumptions of the mock model, the redshift dipole is therefore a highly significant departure from isotropy.

5.3 Residual quadrupole (full mask)

After dipole subtraction, the residual mean-z map shows strong power at $\ell = 2$:

- $C2_{\text{real}} \approx 4.86 \times 10^{-3}$
- $D2_{\text{real}} \approx 4.64 \times 10^{-3}$

The mock suite gives

- $\langle D2_{\text{residual}} \rangle_{\text{mocks}} \approx 4.19 \times 10^{-6}$
- $\sigma(D2_{\text{residual}})_{\text{mocks}} \approx 2.67 \times 10^{-6}$

Formally,

$$z_{\text{D2}} \gg 1000 \sigma$$

This reflects the fact that the mock model produces a residual quadrupole roughly two orders of magnitude smaller than the real one. The exact σ value is not to be taken literally, but the qualitative conclusion is that the residual quadrupole in Quaia is far larger than what these simple mocks can generate.

5.4 Residual quadrupole with $|b| > 30^\circ$

To test whether the quadrupole is dominated by the Galactic plane, we impose $|b| \geq 30^\circ$ in both the real map and mocks. For the real residual map:

- $C2_{\text{real}}(|b| > 30^\circ) \approx 2.00 \times 10^{-4}$
- $D2_{\text{real}}(|b| > 30^\circ) \approx 1.91 \times 10^{-4}$

The mocks (same $|b|$ cut) give:

- $\langle D2_{\text{residual}} \rangle_{\text{mocks}} \approx 2.13 \times 10^{-6}$
- $\sigma(D2_{\text{residual}})_{\text{mocks}} \approx 1.40 \times 10^{-6}$

So

$$z_{D2}(|b| > 30^\circ) \approx (1.91 \times 10^{-4} - 2.13 \times 10^{-6}) / (1.40 \times 10^{-6}) \approx 135 \sigma$$

Even outside the Galactic plane, the quadrupole is still roughly 100× higher than expected from these mocks.

5.5 Quadrupole axis

Using the $\ell = 2$ component of the residual map, the direction of maximum $|\Delta z|$ is found to be:

- $(l_{\text{quad}}, b_{\text{quad}}) \approx (185.6^\circ, 84.2^\circ)$

This is very close to the North Galactic Pole, so the quadrupole pattern is roughly a ring vs poles structure with a symmetry axis almost aligned with the Galactic Z-axis.

Approximate angular separations:

- $\text{sep}(\text{quad}, \text{z-dip}) \sim 70^\circ$
- $\text{sep}(\text{quad}, \text{count dip}) \sim 44^\circ$ (using the count dipole from Quaia counts)
- $\text{sep}(\text{quad}, \text{CMB dipole}) \sim 41^\circ$
- $\text{sep}(\text{quad}, \text{GC}) \sim 96^\circ$
- $\text{sep}(\text{quad}, \text{anti-GC}) \sim 84^\circ$

The quadrupole axis is therefore distinct from the dipole axis yet still part of the broader zoo of large-scale preferred directions.

6. Relation to BAO D_H/D_M Anisotropy

Previous PLamb/FR/STF work has used BAO measurements to probe anisotropic expansion, especially through the ratio D_H/D_M of radial to transverse BAO distances. In some directions and redshift ranges, analyses have suggested $D_H/D_M \approx 0.35$, indicating suppressed radial expansion relative to transverse modes along a preferred axis.

Qualitatively, the Quaia results fit into this picture as follows:

- The redshift dipole in Quaia is a direct signature of directional modulation in the distance–redshift relation.
- The quadrupole indicates that the anisotropy is not purely dipolar: there is a strong two-lobed component, likely encoding more complex angular structure in the STF or in survey systematics.
- Together with BAO DH/DM deviations and HHT-based IMF locking to BAO scales in SN residuals, the Quaia anisotropies strengthen the case for a common underlying large-scale structure (or symmetry breaking) that affects distance measures across multiple tracers.

From a purely phenomenological point of view, Quaia provides an independent quasar-based confirmation that the sky is not perfectly isotropic in redshift space under simple Λ CDM + isotropic $H(z)$ assumptions. Whether this eventually maps onto a rotating STF, variable- c , or some more conventional set of anisotropic or inhomogeneous models remains an open question.

7. Scripts Used (overview and sample calls)

- `quaia_to_csv.py`

Extracts RA, DEC, redshift_quaia from `quaia_G20.0.fits` into a compact CSV.

Example:

```
%run /Users/boyde/.spyder-py3/quaia_to_csv.py \  
--fits /Users/boyde/.spyder-py3/quaia_G20.0.fits \  
--out /Users/boyde/Downloads/quaia_G20p0_basic.csv
```

- qso_add_galactic_coords.py

Adds Galactic coordinates to a CSV with RA, DEC, Z.

Example:

```
%run /Users/boyde/.spyder-py3/qso_add_galactic_coords.py \
--csv "/Users/boyde/Downloads/quaia_G20p0_basic.csv"
```

- quaia_redshift_dipole.py

Builds a mean-z map and fits a redshift dipole (no selection map).

Example:

```
%run /Users/boyde/.spyder-py3/quaia_redshift_dipole.py \
--csv "/Users/boyde/Downloads/quaia_G20p0_basic.csv" \
--nside 64 \
--min-per-pixel 10
```

- quaia_redshift_dipole_sel.py

As above, but includes selection_function_NSIDE64_G20.0.fits and a sel_min_frac threshold to exclude low-completeness pixels.

Example:

```
%run /Users/boyde/.spyder-py3/quaia_redshift_dipole_sel.py \
--csv      "/Users/boyde/Downloads/quaia_G20p0_basic.csv" \
--nside    64 \
--min-per-pixel 10 \
--sel-map  "/Users/boyde/.spyder-py3/selection_function_NSIDE64_G20.0.fits" \
--sel-min-frac 0.3
```

- `quaia_dipole_residual_map.py`

Fits the redshift dipole, builds the dipole model map, subtracts it to get the residual mean- z map, and computes the residual C_ℓ spectrum up to l_{max} .

Example:

```
%run /Users/boyde/.spyder-py3/quaia_dipole_residual_map.py \
--csv      "/Users/boyde/Downloads/quaia_G20p0_basic.csv" \
--nside    64 \
--min-per-pixel 10 \
--sel-map   "/Users/boyde/.spyder-py3/selection_function_NSIDE64_G20.0.fits" \
--sel-min-frac 0.3 \
--out-dipole-map
"/Users/boyde/Downloads/quaia_G20p0_nside64_zdipole_model.fits" \
--out-residual-map "/Users/boyde/Downloads/quaia_G20p0_nside64_zresidual.fits" \
--lmax      10
```

- `quaia_random_mock_zdipole_quad.py`

Generates mock catalogues based on the Quaia random file, fits the redshift dipole, and computes residual quadrupole power D_2 residual for each mock, with optional $|b| > b_{\text{min_abs}}$ cut.

Example:

```
%run /Users/boyde/.spyder-py3/quaia_random_mock_zdipole_quad.py \
--data-csv  "/Users/boyde/Downloads/quaia_G20p0_basic.csv" \
--random-fits "/Users/boyde/.spyder-py3/random_G20.0_10x.fits" \
--sel-map   "/Users/boyde/.spyder-py3/selection_function_NSIDE64_G20.0.fits" \
--nside     64 \
--min-per-pixel 10 \
```

```
--sel-min-frac 0.3 \  
--nmocks      50 \  
--lmax        10 \  
--bmin-abs    30.0 \  
--seed        42 \  
--outcsv      "/Users/boyde/Downloads/quaia_G20p0_randommocks_b30_stats.csv"
```

8. Conclusions and Next Steps

The Quaia $G < 20$ catalogue exhibits a statistically robust redshift dipole and an even stronger quadrupole in the residual mean-redshift field after dipole removal. Under simple random + selection null models, both signals are strong outliers, especially the quadrupole, which remains significant even outside the Galactic plane.

From a PLamb/FR/STF perspective this provides another piece of evidence that the large-scale STF may not be isotropic, and that radial vs transverse distance measures (BAO D_H/D_M , quasar $\langle z \rangle$, SN residuals) all show related anisotropic structure. From a survey-analysis perspective it motivates the construction of more sophisticated mocks and systematics models to test whether any combination of calibration, extinction, and completeness effects could reproduce the observed low- ℓ structure.

Further work could include:

- Extending the analysis to Quaia $G < 20.5$ using the corresponding random and selection files.
- Incorporating more realistic systematics into the mock pipeline.

- Jointly fitting BAO, SN, and Quasar anisotropy within the FR/STF framework to test variable- c or rotating-STF hypotheses quantitatively.